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PRODUCT STRATEGY REPORT

Nutrition Practice SaaS

A Brazil-first, AI-assisted platform for nutritionists and patients

Market benchmark • MVP definition • Architecture • Economics • 16-week execution plan

Decision thesis

Win switchers from legacy nutrition software with a safer migration path, dramatically faster care workflows, a patient adherence loop, and a focused practice-growth workspace. AI is embedded as an auditable copilot; it never autonomously prescribes.

Prepared for Igor Melo

Solo founder • 12–16 weeks to production • Initial operating ceiling R$1,000/month

Research cut-off: 18 August 2026 | Language: English | Product locale: pt-BR first

How to read this report

Evidence labels used throughout:

- Verified: directly observed on an official product, regulator, app-store, or provider source dated at the research cut-off.
- Observed: reported in a public review or community discussion; useful signal, not a representative sample.
- Inference: product opportunity or architectural conclusion derived from multiple sources and the founder constraints.

> **Legal boundary**
> This is a product and engineering strategy, not a legal opinion. Brazilian counsel and a privacy professional should validate data-controller roles, the DPA, lawful bases, retention/deletion conflicts, international transfers, telenutrition consent, and professional-document requirements before production.

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1. Executive Summary Back to TOC

Recommendation in one sentence

Build a modular-monolith SaaS for independent Brazilian clinical nutritionists switching from WebDiet, Dietbox, or similar tools: migration-first onboarding, a fast consultation-to-plan workflow, a patient adherence PWA, and an ethics-aware practice-growth copilot.

The market is large enough for a focused vertical SaaS. The CFN homepage reports 202,903 registered nutritionists in its 2026 second-quarter snapshot. Incumbents are feature-rich and trusted, but current app-store reviews still surface instability, buried logging flows, editing friction, and weak mobile execution. New AI-first entrants prove that automatic diet generation is already becoming a baseline claim rather than a durable moat.

Source: CFN professional statistics — 202,903 nutritionists displayed for Q2 2026

The product wedge should therefore be neither “another diet calculator” nor “ChatGPT for nutritionists.” It should own the complete business and care loop:

- Switch safely: assisted import, validation, an explicit migration report, and exportability from day one.
- Work faster: pre-consult summary → assessment → plan draft → professional approval → publication.
- Improve outcomes: low-friction patient check-ins, substitutions, progress, and an attention queue for the nutritionist.
- Grow the practice: positioning, weekly/monthly content planning, compliant drafts, a public profile/lead form, and follow-up tasks.
- Create trust: no autonomous prescription, clear AI labels, approval gates, provenance, audit logs, and privacy-by-design.

Founder-constrained decision Back to TOC

With one developer, a 12–16 week deadline, fewer than five initial pilot contacts, and a R$1,000 monthly operating cap, microservices would reduce delivery probability without improving the first customer outcome. The recommended backend is a Java 25 + Spring Boot 4 modular monolith, paired with Next.js/TypeScript and PostgreSQL. Domain boundaries and an outbox prepare later extraction, but there is no Kubernetes, Kafka, Redis, GraphQL, or vector database in the MVP.

Expected launch Back to TOC

Milestone	Target	Definition
Prototype validation	End of Sprint 2 / week 6	Five nutritionist interviews; tested migration and plan-builder prototype.
Private beta	End of Sprint 5 / week 12	Core professional and patient loop usable with real pilot data under signed beta terms.
Production MVP	End of Sprint 7 / week 16	Paid switcher onboarding, security/restore drills, observability, support runbook, and launch gates passed.

2. Market Research Back to TOC

Market shape Back to TOC

Brazil has a mature professional-software category: WebDiet and Dietbox combine prescription, patient apps, scheduling, education, and communication; smaller products compete on price or AI; adjacent tools address WhatsApp conversion, social content, or no-code paid programs. International platforms show that the category can expand from meal planning into an operating system for a health practice.

Signal	Verified evidence	Strategic meaning
Large professional base	CFN displays 202,903 nutritionists (Q2 2026).	A narrow initial segment can support a meaningful SaaS without serving every nutrition specialty.
Incumbent scale	Dietbox patient app: 1M+ Google Play downloads and 119K reviews; WebDiet patient app: 1M+ downloads.	Switchers expect a mature patient experience and cannot tolerate lost records or downtime.
AI substitution pressure	CFN warned in Jan 2026 about illegal plans and nutrition guidance by unqualified people; current reporting describes consumers using general chatbots for diets.	Position the product as the professional-controlled alternative: personalization, accountability, and safety.
Practice growth is fragmented	Dietbox integrates Canva and practitioner discovery; WebDiet sells education/marketing; specialist tools automate content or WhatsApp funnels.	Growth is a real job-to-be-done, but a full marketing suite would be an MVP trap.
AI is already crowded	WebDiet, Dietbox, NutriAssist, Cibus, PrescrIA Pro, and other emerging products market AI-assisted plans or summaries.	Differentiation must be workflow quality, evidence, governance, adherence, and migration—not model access.

Source: Dietbox Google Play

Source: WebDiet Google Play

Source: CFN warning on unqualified digital nutrition guidance

Research limitations Back to TOC

Competitor pages use promotions, billing toggles, and geo-specific offers; prices below are snapshots, not promises. Public reviews over-represent strong experiences, and vendor claims were not penetration-tested or clinically evaluated. Emerging 2026 websites sometimes make unverified adoption, security, or accuracy claims; this report treats them as messaging evidence only.

3. Competitor Benchmark Back to TOC

Brazil: major and emerging platforms Back to TOC

Product	Audience / model	Price snapshot	Professional / patient experience	Growth & AI	Evidence status
WebDiet	Nutritionists; individual plans	Premium: R$49.90/mo first 3 months, then R$94.90; Black R$139.90	Broad prescription methods, TACO/TBCA/manufacturer foods, patient app, diary, goals, scheduling	Clara AI on Black; courses, Canvas, WhatsApp/Google Calendar	Official, 18 Aug 2026
Dietbox	Nutritionists/students; subscription	Promo R$49.90/mo first 3 months; subsequent price displayed around R$88–92 depending selector	Professional + patient apps, USDA/food tables, plans, anthropometry, diary, chat, appointments	Assistant, WhatsApp, Canva, discovery, Google Calendar	Official; price UI variable
NutriAssist	AI-first Brazilian nutritionists	R$99.90 monthly; R$79.90/mo annual equivalent; credit packs	Clinical chat, diet editor, TACO/TBCA/USDA, exams, forms, calendar, WhatsApp bot	Strong contextual AI/RAG and workflow messaging	Vendor-verified feature/pricing claims
Salutes	Health practices; all-in-one	Promotional page: R$97/mo	Agenda, chart, plan, assessments, finance, CRM, WhatsApp, patient app	AI writing/assistant/lead bots	Vendor claims; limited independent review
Avanutri Online	Nutritionists; protocol depth	R$130/mo official store result; six-month trial claimed	Extensive protocols/indicators, food base, substitutions, patient diary/chat	Less differentiated AI/growth story	Official/vendor claims; food count inconsistent
CoreDiet / niche entrants	Emerging Brazilian products	Not consistently published	Clinical suite, patient app, specialist modules	CorePage/Marketing AI; several entrants claim AI plans/photos	Treat adoption/security claims as unverified

Source: WebDiet pricing

Source: Dietbox product and pricing

Source: NutriAssist pricing and features

Source: Salutes

Source: Avanutri Online

International benchmarks Back to TOC

Product	Price snapshot	Strengths worth learning from	Gap / caution for Brazil
Healthie	Core US$19.99; Essentials US$49.99; Plus US$129.99; Group US$149.99+	Practice OS: scheduling, payments, charting, telehealth, client portal/mobile, organizations	US healthcare/billing orientation; broad scope and higher complexity
Practice Better	Free 3 clients; US$35 Starter; US$59 Professional; US$89 Plus; US$145 Team	Strong programs, journals, goals, forms, mobile, team operations; metered AI charting	Food planning partly depends on higher tiers/integration; pricing jump complaints appear in reviews

Product	Price snapshot	Strengths worth learning from	Gap / caution for Brazil
That Clean Life	US$30 Starter; US$60 monthly or US$35 annual-equivalent Plus	Beautiful plan delivery, templates, filters, dietary constraints, nutrition insights	Narrower practice management; no free trial
NutriAdmin	Approx. US$34.99 Basic; US$49.99 Popular; US$74.99 Pro monthly	CRM, portal, questionnaires, reports, meal planning, Google Calendar	AI recipe add-on/feature segmentation; international food expectations
Nutrium	Current official public pricing unclear; a Mar 2026 third-party screenshot reported US$15–25 annual-equivalent	Integrated professional workflow and patient follow-up across many countries/languages	Pricing/status uncertainty; do not base economics on third-party snapshot

Source: Healthie pricing

Source: Practice Better pricing

Source: That Clean Life pricing

Source: NutriAdmin pricing

Source: Nutrium review with observed March 2026 pricing — Third-party competitor; use cautiously

Patient/mobile evidence Back to TOC

Observed source	Finding	Product implication
Dietbox Google Play, Jun 2026	A reviewer described meal logging as buried under four UI levels, leading to delayed batch completion.	Make Today the default patient screen; one-tap check-in and substitution, not a generic dashboard.
Dietbox App Store, 2025–26	Reviews mention interface/UX problems, instability, editing problems, and login issues; updates frequently address diary/plan/login.	Reliability and task completion are marketable features. Track crash-free sessions and completion time.
WebDiet Google Play, Aug 2026	A reviewer requested direct recipe access from the meal instead of leaving the plan and searching a large list.	Every plan object should be contextual and deep-linked—recipe, substitution, rationale, and logging.
WebDiet Google Play, Aug 2026	A reviewer requested supplement/GI/Bristol-style longitudinal logging and symptom correlations.	Post-MVP opportunity: configurable clinical check-ins rather than universal calorie logging.
Dietbox professional App Store, 2025	Visible reviews report disappearing plan options, editing failures, and multi-day impact on care.	Autosave, explicit version history, idempotency, and restore testing are product differentiation.

Source: Dietbox professional App Store reviews

Market expectations, advantages, and gaps Back to TOC

Category	Capabilities
Must-have expectations	Patient records; assessment/anamnesis; measurements; calculated/free plans; food database; substitutions; recipes or meal guidance; patient plan access; progress; reminders; basic communication; export/PDF.

Category	Capabilities
Stronger-competitor advantages	WhatsApp/Calendar/Canva; mobile apps; templates; courses; team roles; telehealth; finance; automated forms; structured programs; protocol libraries; AI summaries/plans.
Repeated gaps	Complex or fragile UX; plan creation still time-consuming; logging friction; context lost between plan/recipe/substitution; adherence signals not actionable; communication fragmented; mobile reliability; opaque migration/export; AI trust.
White-space opportunity	Switcher-grade migration + time-to-plan leadership + adherence attention queue + ethics-aware practice growth + auditable AI approvals.

4. User Pain Points Back to TOC

Persona / job	Pain	Underlying cause	Design response
Switcher nutritionist	Fear of losing records, templates, and continuity	Weak exports, proprietary formats, manual re-entry	Import workspace, mapping preview, reconciliation report, rollback, parallel-read period, permanent export.
Nutritionist / consultation	Too much work before and after the appointment	History scattered; repeated typing; slow plan construction	Pre-consult brief, structured note draft, reusable protocols, plan versioning, keyboard-first editor.
Nutritionist / growth	Inconsistent demand and difficulty turning expertise into clients	No weekly system; content and leads live in separate tools	Practice-growth cockpit: positioning, content calendar, drafts, lead form, follow-up tasks.
Nutritionist / monitoring	Cannot tell who needs attention	High-volume diaries and messages without prioritization	Attention queue based on missed check-ins, questions, deviations, and upcoming follow-ups.
Patient / daily use	Logging feels like work; plan is hard to navigate	Generic dashboards, deep menus, calorie-centric design	Today-first PWA, one-tap adherence, contextual substitutions/recipes, optional detail.
Patient / questions	General AI is instant but unsafe and ignores the professional plan	No plan-grounded, 24/7 explanation layer	Read-only plan assistant with approved sources, clear escalation, and no autonomous modifications.

5. Market Opportunities Back to TOC

1 — Migration as a product

Targeting switchers changes onboarding from a form into a trust event. Offer assisted import, evidence of what migrated, and a no-lock-in export pledge.

2 — Practice growth tied to clinical retention

Do not sell vanity content volume. Connect positioning → content → lead → consultation → ongoing care → renewal, while respecting CFN advertising rules.

3 — Adherence intelligence

Convert lightweight patient check-ins into a small, explainable attention queue. The nutritionist acts on exceptions instead of reading every event.

4 — Professional-controlled AI

General chatbots compete on immediacy. The platform competes on context, professional accountability, auditability, safety gates, and follow-through.

5 — Brazilian food and routine reality

TACO/TBCA provenance, household measures, rice/beans, restaurant-by-weight, budget and preparation constraints, and weekend patterns matter more than an enormous undifferentiated food catalog.

6. Product Differentiation Strategy Back to TOC

Positioning

The modern practice OS for Brazilian nutritionists who want to switch software, save clinical time, keep patients engaged, and grow ethically—without surrendering judgment to AI.

Pillar	Promise	Proof metric	MVP expression
Switch without fear	Your history and templates come with you—and can leave again.	% imports reconciled; migration defects; time to first usable patient	CSV/PDF import, mapping, exceptions, export bundle
From consultation to plan quickly	Finish the work while the context is fresh.	Median time from consultation end to published plan	Structured assessment, fast plan builder, draft summary, approval
Know who needs you	One attention queue, not an ocean of logs.	% flagged patients acted on; adherence response time	Check-ins, missed-event rules, explainable flags
Grow with integrity	Turn expertise into a repeatable pipeline.	Content plan completion; lead-to-consult conversion	Positioning brief, calendar, compliant drafts, lead form/tasks
AI under professional control	Every clinical output is traceable and approved.	AI acceptance/edit/rejection; safety incidents	Structured outputs, labels, provenance, audit, publish gates

Practice-growth scope boundary Back to TOC

MVP includes strategy and workflow, not a social network scheduler. It creates a monthly theme, weekly plan, evidence-backed draft, caption variants, short-video outline, and call-to-action; checks the draft against a ruleset; stores approval status; and connects a public profile/lead form to follow-up tasks. It does not auto-post, scrape social networks, buy ads, promise outcomes, display public fees/promotions, or generate before/after content.

Source: CFN Code of Ethics, Resolution 599/2018

Source: CRN-8 social-media and marketing FAQ — Public prices/promotions/sweepstakes and before/after guidance

7. AI Strategy Back to TOC

Principles Back to TOC

- Deterministic first: calculate nutrients, constraints, allergens, and totals with code and curated data—not an LLM.
- Grounded second: patient answers use the active professional-approved plan and approved educational content only.
- Structured outputs: JSON Schema for every clinical draft; refuse or escalate when the schema/safety policy fails.
- Human in the loop: the nutritionist approves any item that changes care, becomes part of the record, or is sent as clinical guidance.
- Minimum necessary data: pseudonymize where practical; do not include identity fields when not needed; redact logs.
- Auditable by design: model snapshot, prompt/policy version, input references, output, cost, latency, safety flags, reviewer, and final edits.
- Provider-portable, not provider-agnostic fantasy: one production provider behind an internal interface; add a second only when risk or economics justify it.

Capability ranking Back to TOC

Score = 30% user value + 25% differentiation + 15% implementation ease + 10% low operating cost + 20% low safety risk. Scores are strategic estimates (1–5), not empirical results.

Rank	Capability	Value	Diff.	Ease	Cost	Safety	Score	Decision
1	Pre-consult/ history summary	5	4	5	5	4	4.55	MVP
2	Notes → structured consultation draft	5	4	4	5	4	4.40	MVP, typed notes
3	Adherence digest + explainable flags	5	5	3	4	4	4.35	MVP rules + summary
4	Ethics-aware content planner/drafts	4	5	4	4	4	4.25	MVP-lite
5	Plan-grounded patient Q&A	5	5	3	4	2	3.95	Controlled beta
6	Smart substitutions	5	4	3	4	3	3.90	Rules first; AI explanation

Rank	Capability	Value	Diff.	Ease	Cost	Safety	Score	Decision
7	Meal-plan draft from constraints	5	3	3	3	2	3.35	MVP professional-only
8	Audio consultation scribe	4	3	2	3	3	3.05	Post-MVP
9	Meal photo nutrient estimation	4	3	2	2	1	2.45	Avoid initially
10	Autonomous patient coach/prescriber	4	4	1	2	1	2.40	Do not build

Approval policy Back to TOC

AI output	Professional approval	Patient exposure	Guardrail
History/pre-consult summary	Review before consultation; corrections logged	No	Citations to source fields; missing-data flags
Structured consultation note	Required before record finalization	No	Draft watermark; immutable final version
Nutrition-plan draft	Required before publish; never bulk auto-publish	Only approved version	Constraint engine, allergen/restriction validation
Substitution inside pre-approved group	Rule-level pre-approval; nutritionist defines group	Yes	Nutrient tolerance, restrictions, serving conversion
Substitution outside group	Case-by-case required	Not until approved	Escalate; no invented foods/portions
Patient plan explanation	No per-message approval if within approved plan	Yes, labeled AI	Retrieval allowlist; no diagnosis/change; escalation
Lab/supplement interpretation	Always required	Not MVP	Professional-only, source and uncertainty displayed
Social content	Professional approval before copy/export	Public after approval	CFN ruleset, evidence links, no claims/fees/before-after

Model and operational design Back to TOC

Start with the OpenAI Responses API behind an internal AiGateway interface. Use a cost-sensitive model for patient explanations/classification and a stronger balanced model for professional summaries/drafts; route by task, not by a user-facing model picker. As of 18 Aug 2026, OpenAI lists GPT-5.6 Luna at US$0.20/US$1.20 and Terra at US$2/US$12 per 1M input/output tokens. Use model snapshots and a task-level budget.

OpenAI states that API data is not used for training unless the customer opts in, but default abuse-monitoring logs may be retained for up to 30 days. Set store:false, complete a DPA and international-transfer review, minimize health data, and assess Modified Abuse Monitoring or Zero Data Retention eligibility. Do not market “zero retention” unless the account and endpoint configuration actually provides it.

Source: OpenAI model catalog and pricing
Source: OpenAI data controls
Source: OpenAI Structured Outputs

Evaluation and prompt-injection defenses Back to TOC

- Golden sets: 100+ de-identified cases covering allergies, diabetes, pregnancy, renal risk, eating-disorder signals, pediatric cases, cultural constraints, and ambiguous notes.
- Measure: schema validity, groundedness, constraint violations, omission rate, unsafe recommendation rate, nutritionist edit distance, acceptance/rejection, latency, and cost.
- Prompt injection: treat uploaded documents and patient messages as untrusted data; separate instructions from content; tool allowlists; no arbitrary URL/browser tools; escape/quote retrieved text; output policy after generation.
- Red-team before every material prompt/model change; version and canary; disable a feature independently through flags.
- Patient assistant must have an emergency/clinical escalation path and clearly state when the nutritionist—not the AI—must respond.

8. MVP Definition Back to TOC

MVP outcome Back to TOC

A switching nutritionist can import a small real practice, complete an assessment, publish a safe plan, let a patient follow/check in from a mobile web app, see who needs attention, and run a simple weekly growth routine. Everything else is subordinate.

MVP module	Included	Explicitly not included
Switcher onboarding	Organization + professional profile; manual CRN review; CSV patient import; PDF/document batch; mapping preview; exception report; export	Automatic scraping of competitor accounts; every proprietary format; white-glove migration at scale
Patient/clinical record	Patient profile, consent, intake/anamnesis, core history, consultation note, goals, measurements, progress photos	Full EHR, every protocol, lab interpretation, prescriptions for all specialties
Nutrition plan	TACO-seeded food search with provenance; meals/items/household measures; totals; templates; versioning; substitutions; PDF	TBCA bulk ingest without license; barcode catalog; recipe marketplace; exotic optimization engine
Patient PWA	Invitation, Today, plan, contextual substitution, one-tap adherence, measurements/photos, questions, reminders	Native app, wearables, offline conflict engine, calorie-photo estimation
Adherence	Configurable check-in, missed-check-in rules, attention queue, simple progress graphs	Predictive medicine, gamification economy, generalized behavioral scoring
AI	History summary, typed-notes structuring, professional plan draft, adherence digest, controlled plan Q&A beta	Autonomous diagnosis/prescription, lab/supplement advice, audio scribe, open-web patient bot
Practice growth	Positioning profile, monthly/weekly content planner, AI drafts + ethics checklist, public profile/lead form, follow-up tasks	Auto-posting, ad buying, social inbox, full CRM, WhatsApp sales bot

MVP module	Included	Explicitly not included
Operations	Basic appointment record + email reminders; support/admin; audit; analytics; manual subscription administration	Telehealth video, integrated payment split, insurance, finance/accounting

MVP acceptance guardrails Back to TOC

- A new switching nutritionist reaches a usable imported patient in ≤30 minutes for the supported template.
- A trained user creates and publishes a standard plan in ≤15 minutes excluding clinical reasoning and consultation time.
- A patient finds today's meal and records adherence in ≤10 seconds from opening the PWA.
- No clinical AI output reaches a patient unless the approval policy permits it; plan changes always require professional approval.
- Tenant-isolation, consent, export, deletion workflow, audit, backup restore, and incident runbook tests pass before real data.

9. Post-MVP Roadmap Back to TOC

Horizon	Capabilities	Why then
Post-MVP / retention	Native Expo patient app; secure asynchronous messaging; flexible check-ins; recipes/shopping; Google Calendar; better migrations; renewal/reactivation; content performance/manual attribution	Adds retention after the core care loop and migration wedge are proven.
Growth / PMF	Clinic roles; reception; multi-professional teams; WhatsApp Business integration; billing/PIX; audio scribe; lab extraction; protocol knowledge base; TBCA commercial agreement; branded programs	Requires volume, support capacity, legal review, and proven willingness to pay.
Advanced	Wearables; population cohorts; experiments; richer AI routing; enterprise SSO; APIs/webhooks; read replicas; regional expansion; specialty modules	Only after repeatable acquisition and strong retention.
Avoid initially	Microservices, Kubernetes, Kafka, a proprietary general food database, autonomous AI care, meal-photo calorie promises, marketplace, insurance billing, video infrastructure, full social scheduler	High complexity, regulation, cost, or weak validation relative to the 16-week objective.

10. Recommended Technology Stack Back to TOC

Layer	Recommendation	Why this product	Rejected for MVP
Web	Next.js App Router + TypeScript; responsive PWA	Fast React delivery, routing/rendering flexibility, accessibility, future shared TS packages	Separate web products; heavy micro-frontends

Layer	Recommendation	Why this product	Rejected for MVP
Backend	Java 25 LTS + Spring Boot 4.1 + Spring Modulith	Matches founder expertise; strong security/validation/transactions/observability; explicit module boundaries	NestJS is viable but offers insufficient advantage to discard Spring depth
Mobile	Expo/React Native after MVP	Best fit with React/TS; first-class monorepo support; share tokens, API client, schemas, i18n—not whole UI	Flutter/native teams before PMF
Database	PostgreSQL; Flyway migrations	Transactional clinical model, JSON where useful, constraints, mature portability	MongoDB as primary; database-per-tenant
API	REST + OpenAPI 3; generated TS client; webhooks later	Clear resource workflows, caching/idempotency, easy mobile/integration support	GraphQL adds authorization/caching/operational surface without a current need
Auth	Supabase Auth; PKCE; own authorization tables; TOTP MFA	Managed identity, São Paulo region option, mobile/web support, low MVP cost	Self-hosted Keycloak/Auth server; business rules in provider claims
Files	Supabase Storage initially, private buckets + signed URLs; offsite object inventory/backup	Co-located managed data plane; simple client uploads	Database BLOBs; public buckets
Jobs	Transactional outbox + PostgreSQL-backed jobs/JobRunr	Reliable email, AI, export, import without a new broker	Kafka/SQS/Redis queue before load requires it
Cache/search	None initially; PostgreSQL indexes and full-text/trigram	Avoid invalidation and extra infrastructure	Redis/vector/search service without measured need
AI	OpenAI Responses through AiGateway; Structured Outputs; task router	Current strong multilingual model range and governance primitives	Hard-coded vendor calls across domains
Observability	OpenTelemetry/Micrometer + Sentry; sanitized product events	One trace ID from UI/API/job/AI; low-cost free tiers	PHI in logs/session replay

Source: Spring Boot official project
Source: Oracle Java SE support roadmap — Java 25 is LTS
Source: Next.js App Router docs
Source: Expo monorepo support

11. System Architecture Back to TOC

Logical architecture Back to TOC

Client edge	Application core	Managed data/services
Nutritionist web • Patient responsive PWA • Future Expo mobile • Future integrations	REST API • AuthZ policy • domain modules • outbox/jobs • AI gateway • audit/analytics events	PostgreSQL/Auth/Storage • email • OpenAI • Sentry/metrics • CDN/DNS

Request path: client obtains an identity token → backend validates identity and resolves organization/membership → domain service enforces relationship/role and performs one transaction → audit/outbox rows commit atomically → worker performs notification/AI/export → result is stored and emitted as a sanitized product event.

Modular monolith boundaries Back to TOC

Module	Owns	May depend on
Identity & Tenancy	User, organization, membership, role, invitation, care-team access	Auth provider adapter
Growth	Professional profile, positioning, content plan/draft, lead, follow-up task	Identity; Scheduling; AI
Patient	Person, organization-patient relationship, intake, consent, goals	Identity; Audit
Clinical	Consultation, note versions, assessment, measurement, progress media	Patient; Documents
Nutrition	Food provenance, portions, recipes later, plan/version/meal/item/substitution	Patient; Clinical
Engagement	Check-ins, adherence events, attention flags, patient questions	Patient; Nutrition; Notification
Scheduling	Appointment and reminder state	Patient; Notification
AI Governance	Interaction, policy/prompt/model version, evaluation, review decision, cost	Read-only projections from allowed modules
Documents	Object metadata, signed access, export jobs	Identity/Tenancy
Audit & Compliance	Append-only audit, data request/case, retention holds	Receives events; no domain writes

Multi-tenancy Back to TOC

MVP tenancy is shared database/shared schema. Every professional receives an Organization even when solo; clinics later add memberships. Organization-owned tables carry organization_id, and unique/index definitions include it. Application checks are mandatory; PostgreSQL row-level security is defense in depth for exposed Supabase paths. Never trust a tenant ID supplied by the client without resolving it from the authenticated membership.

Option	Strength	Weakness	Decision
Shared DB/schema	Lowest cost; simplest migrations/reporting; best solo velocity	Isolation depends on rigorous policy/tests	MVP recommendation
Shared DB/separate schemas	Some logical separation	Migration explosion; awkward pooling/analytics	Do not use
Separate database	Strong isolation/custom residency	High cost and operational burden	Enterprise option after PMF

Authentication and authorization Back to TOC

- Professional: email/password or passwordless with verified email; require TOTP MFA before access to real patient data; step-up for exports and sensitive admin actions.

- Patient: invitation-bound signup with PKCE; short access token, rotating refresh token, device/session view, revocation; no patient enumeration.
- RBAC + relationship policy: OWNER, NUTRITIONIST, ASSISTANT (future), PATIENT, SUPPORT_BREAK_GLASS. Roles do not replace patient-care relationship checks.
- Break-glass support access disabled by default, time-limited, reason-required, fully audited, and visible to organization owner.
- MFA recovery is a product flow: verified recovery, cooldown, audit, and alerts—not a database edit by support.

12. Data Architecture Back to TOC

Core relationships Back to TOC

Aggregate / entity	Key relationships and invariants
Organization	Owns practice data. Solo professional is still a one-member organization. Future clinic staff join through Membership.
User / ProfessionalProfile	User is identity; profile stores CRN region/number/verification. No clinical data in auth claims.
PatientPerson / CareRelationship	A person can have separate relationships with multiple organizations. Records do not merge across organizations without a defined consent/legal process.
ConsentRecord	Purpose, text/version, lawful-basis context, channel, timestamp, withdrawal, guardian where applicable. Consent is not used as a universal legal basis.
Consultation / ClinicalNoteVersion	Consultation anchors assessment. Final notes are immutable; amendments create linked versions with author/reason.
Measurement / Goal / ProgressMedia	Time-series observations with method, unit, source, author; media in private storage with metadata in DB.
NutritionPlan / PlanVersion	Draft → review → published → superseded. Patient sees only published versions. Plan item references FoodVersion/portion, not mutable display text alone.
Food / FoodVersion / FoodSource	Canonical item plus source/version/provenance, nutrient values and qualifiers, household portions, aliases, branded/custom scope.
AdherenceEvent / CheckIn / AttentionFlag	Raw events are separate from derived flags. Every flag records rule/model, evidence, severity, status, and resolution.
ContentPlan / ContentDraft / Lead	Growth data is separated from clinical data. A lead becomes a patient only through an explicit conversion and consent flow.
AIInteraction / AuditEvent	AI stores task/policy/model/input references/output/review/cost; audit stores security-sensitive state changes. Neither is a general log dump.

Food data strategy Back to TOC

TACO 4th edition is a useful open seed: the official publication permits total or partial reproduction when the source is cited, but it is dated (2011) and contains roughly 597 foods. TBCA is much richer and current (version 7.3 at research time; its site describes more than 5,700 foods in the preceding release and Brazilian recipes/household measures), but public search access is not the same as a commercial bulk-data/API license. Do not scrape TBCA into production without written permission or a legal/licensing conclusion.

Source: TACO 4th edition PDF

Source: TBCA official site

Source: FAO TBCA catalog entry

Stage	Data approach	Controls
MVP	TACO ingestion after legal confirmation; custom foods; nutritionist-scoped foods; household portions; source/version visible	Provenance required; no silent value override; import tests; citation in exports
Post-MVP	Written TBCA license/permission; USDA for international foods; branded source partnership or carefully licensed Open Food Facts	Per-source license registry; locale/brand confidence; duplicate resolution; moderation
Scale	Canonical matching, source precedence, version migrations, validation sampling, issue reporting	Stewardship workflow; change audit; backward reproducibility for old plans

13. Security & LGPD Strategy Back to TOC

Legal/regulatory requirements vs recommendations Back to TOC

Type	Requirement / finding	Product or engineering action	Validation
Law	Health data is sensitive personal data under LGPD Art. 5; Art. 11 has specific lawful bases.	Data inventory, purpose/basis register, minimum necessary collection; do not use legitimate interest as a generic sensitive-data basis.	Brazil privacy counsel
Law	Data-subject rights include confirmation, access, correction, portability, information, revocation, and deletion subject to legal retention.	Privacy request case workflow; export; correction; identity verification; legal hold/retention reason; response tracking.	Counsel + privacy lead
Law	Security-by-design and incident duties apply; ANPD Resolution 15/2024 requires notice within three business days when relevant risk/damage exists.	Incident classification/playbook, evidence preservation, controller notification SLA, contact registry, tabletop exercise.	Counsel; controller DPA
Regulation	Telenutrition requires active CRN, e-Nutricionista registration, information on limitations/fragilities, and consent before first remote service under CFN 760/2023.	Professional attestation, consent/version evidence, modality field, document/signature workflow if teleconsultation enters scope.	CFN specialist/counsel
Regulation	CFN record rules and Law 13.787 establish long retention; CFN 594 specifies at least 20 years for physical records and rules for electronic records.	Retention schedule and legal holds; never equate account cancellation with clinical-record deletion; immutable final-note versions.	Counsel; CRN interpretation

Type	Requirement / finding	Product or engineering action	Validation
Regulation	CFN advertising rules prohibit misleading/sensational claims, result guarantees, public fees/promotions/sweepstakes, and before/after use.	Content checklist and warnings; templates avoid claims; approval; evidence link; no automated publishing in MVP.	CFN/CRN marketing review
Law	International transfers must use LGPD mechanisms under ANPD Resolution 19/2024.	Vendor register, data-flow map, SCC/contract mechanism, subprocessor review, disclosure and transfer-impact review.	Brazil privacy counsel
Best practice	Strong access, encryption, backup and monitoring are expected even where a precise configuration is not legislated.	TLS; managed encryption at rest; KMS/provider keys; MFA; least privilege; secret manager; SAST/DAST; restore drills.	Security review

Source: Brazilian LGPD, Law 13.709/2018

Source: ANPD security-incident communication

Source: ANPD international-transfer guidance

Source: CFN telenutrition Resolution 760/2023 announcement

Source: CFN Resolution 594/2017

Source: Law 13.787/2018 on health-record digitization/retention

Controller/operator posture Back to TOC

Likely structure: the nutritionist/clinic is controller for patient care data and the SaaS is operator; the SaaS is controller for its own accounts, billing, fraud/security, and possibly carefully defined product analytics. Roles can vary by purpose, so contracts and notices must be purpose-specific. Execute a DPA with instructions, subprocessors, security annex, incident obligations, international transfers, return/deletion, audit cooperation, and controller response support.

Security baseline before beta Back to TOC

- Threat model tenant isolation, IDOR/BOLA, invitation takeover, object URL leakage, mass export, prompt injection, support access, and malicious file uploads.
- Encrypt in transit; provider encryption at rest; signed short-lived file URLs; virus/type/size scanning; private buckets; no public patient media.
- Least-privilege database/service accounts; production separated from development; no real data in non-production; secrets in managed store; dependency and container scanning.
- Append-only audit for login/MFA changes, role/membership, consent, record read/export, plan publish, note finalization/amendment, AI review, and support access.
- Logs and analytics contain stable pseudonymous identifiers—not names, notes, prompts, plan text, photos, or tokens. Disable session replay on clinical surfaces unless comprehensively masked and legally reviewed.
- Daily managed DB backup plus weekly tested logical export; object inventory and replication/export; documented RPO/RTO; quarterly restore drill initially.

14. Infrastructure & Scalability Strategy Back to TOC

Area	Stage 1 — Early MVP	Stage 2 — PMF	Stage 3 — Scale
Scale	10s–100s professionals; 100s–1,000s patients	1,000s–10,000s users	100,000s–millions
Compute	One Spring app/worker deployment on Fly.io GRU; min 1 instance, health restart; Next.js on Vercel Pro	2+ API instances; separate worker process; autoscaling/load balancer	Multi-AZ app pools; extract only measured hotspots; regional strategy
Database	Supabase Pro PostgreSQL in São Paulo; pooled connections; daily backups	Larger compute; PITR if economics justify; read replica only for measured read load	HA/PITR mandatory; replicas; partition heavy event/audit tables; enterprise tenant options
Storage/CDN	Private Supabase Storage; signed URLs; Vercel/CDN for public assets	Lifecycle policies; object replication/export; malware pipeline	Multi-region object strategy; CDN; media processing workers
Cache	None	Small Redis only for proven hot/cache/rate-limit use	Distributed cache if measured; strict invalidation ownership
Events/jobs	Postgres outbox + JobRunr	Managed queue (SQS/Pub/Sub) when throughput/reliability warrants	Partitioned queues/events; Kafka only for demonstrated streaming needs
Monitoring	Sentry, OTel/Micrometer, uptime, structured redacted logs, cost alerts	Central metrics/traces, SLOs, on-call rotation, synthetic journeys	Multi-region SLOs, capacity forecasting, security monitoring
DR	RPO 24h / RTO 8h target; quarterly restore test	RPO ≤1h / RTO ≤2h; PITR; semiannual failover	RPO minutes / RTO <1h where business requires; regional drills
CI/CD/IaC	GitHub Actions; test/build/scan/migrate/deploy; protected prod; small Terraform footprint	Progressive deploy/canary; ephemeral previews with fake data	Policy-as-code, multi-account/environment, automated rollback

Source: Supabase regions — São Paulo sa-east-1 available

Source: Supabase pricing — Pro US$25/month; daily backups 7 days

Source: Fly.io regions — São Paulo GRU supports Machines and Managed Postgres

Source: Fly.io resource pricing

Source: Vercel pricing — Pro US$20/month

Provider decision Back to TOC

Use a deliberately small multi-provider setup: Vercel for web delivery/previews; Fly.io GRU for the Java workload; Supabase São Paulo for PostgreSQL/Auth/Storage. This is simpler and cheaper for a solo founder than a full AWS production stack in São Paulo. Isolate each provider behind standards: OpenAPI, OIDC/JWT adapter, PostgreSQL/Flyway, object-storage adapter, containers, and Terraform. Revisit consolidation when compliance procurement, private networking, SLAs, or R$10k+ MRR justify it.

15. Estimated Infrastructure / AI Cost Model Back to TOC

Planning exchange rate: US$1 = R$5.20 on 18 Aug 2026; budgets below add approximately 15% FX/tax/variance headroom. Provider taxes and regional compute markups can differ. Re-price before purchase.

Source: USD/BRL reference — About R$5.20 on 18 Aug 2026

Cost item	Launch assumption	USD/month	BRL planning range
Supabase Pro	1 production project, micro compute credit, daily backups	25	R$130–155
Vercel Pro	1 developer; commercial production	20	R$104–125
Fly.io app	1–2 GB shared CPU in GRU; region/egress variance	15–25	R$78–155
Email	Resend free ≤3,000/mo, then Pro if required	0–20	R$0–125
Sentry/analytics/uptime	Free tiers; no clinical payloads	0–10	R$0–62
Backups/storage/domain/misc.	Small object volume, domain, offsite export	5–15	R$26–94
AI	Hard task budgets; pilot to early paid usage	20–70	R$104–437
Total	Expected early range	85–185	Approx. R$500–950 including contingency

Source: Resend pricing — Free 3,000 emails/month; Pro US$20

Source: Sentry pricing

AI unit economics Back to TOC

Task	Assumed tokens	Model class	Approx. cost/action	Control
Patient plan Q&A	3k input / 300 output	Luna	~US$0.001	20/month/patient soft quota; cache approved FAQ
History/adherence summary	6k input / 1k output	Terra	~US$0.024	Generate on state change/consult prep, not every view
Clinical note structure	8k input / 1.5k output	Terra	~US$0.034	One draft + one explicit regenerate
Plan draft	10k input / 2k output	Terra	~US$0.044	Professional-only; deterministic validation; max context
Content package	5k input / 1.5k output	Terra	~US$0.028	Monthly themes, weekly batches, evidence reuse

At 20 professionals using 100 Terra-class actions each plus 600 patients using 20 Luna-class questions, rough AI spend is about US$60–90 (R$312–468) before overhead. That makes quotas, prompt compression, caching, and action-triggered generation necessary. Expose remaining usage clearly and stop at a hard monthly budget rather than creating a surprise invoice.

16. Business Model & Pricing Strategy Back to TOC

Initial ICP Back to TOC

The first commercial segment is a switcher, not a beginner: an independent Brazilian clinical nutritionist currently paying for WebDiet, Dietbox, Nutrium, or a similar product; approximately 20–150 active patients; dissatisfied with speed, stability, patient engagement, AI trust, or practice growth; willing to run a parallel test. Small clinics are a later expansion, not the MVP design center.

Launch offer Back to TOC

Offer	Price	Includes	Reason
21-day switch trial	R$0; no card	Supported import preview, 10 active patients, limited AI, patient PWA	Two weeks is short for a real clinical cycle; import lowers evaluation risk.
Founding switcher	R$79/month for 12 months; first 30–50	One professional, 75 active patients, 300 AI units, assisted migration	Fast learning and testimonials without a permanent free tier.
Public Solo	R$99 monthly or R$89 annual-equivalent	One professional, 75 active patients, core/growth, 300 AI units	Sits near incumbent spend while selling migration + workflow value.
Practice Growth	R$149 monthly post-MVP	150 active, larger AI allowance, advanced growth/automation	Monetizes commercial value and AI usage after validation.
Clinic	From R$299 + seats, post-MVP	Roles, reception, shared care, governance/reporting	Do not absorb clinic complexity into the solo plan.

AI unit should be a transparent task allowance, not opaque tokens. Never limit historical records: a patient limit tied to record deletion conflicts with continuity and retention obligations. Limit active care relationships and monthly AI usage instead. Patients pay nothing initially; premium patient subscriptions would create channel conflict and safety complexity.

Unit economics and break-even Back to TOC

Paid solo accounts	MRR at R$99	Operating estimate	Contribution before tax/support
10	R$990	R$500–750	R$240–490
20	R$1,980	R$650–950	R$1,030–1,330
50	R$4,950	R$1,200–2,000	R$2,950–3,750

These are planning estimates before payment fees, tax, founder labor, customer support, refunds, and legal/accounting costs. The important control is gross-margin telemetry by tenant and AI task from day one.

17. UX / Core User Journeys Back to TOC

Nutritionist switcher journey Back to TOC

Step	Experience	Click/time reduction
1. Evaluate	Interactive demo with fake patient; compare migration coverage; export pledge	No signup before value proof
2. Import	Upload supported CSV/PDF → map → preview → validate → import report	One guided flow; reversible before commit
3. Configure	Professional profile, positioning, plan defaults, brand voice, consent templates	Progressive setup; defaults from import
4. Prepare	Agenda + pre-consult brief + missing intake items	One consultation workspace; no tab hunting
5. Consult	Assessment, notes, measurements, goals with autosave/keyboard flow	No duplicate fields; shortcuts/templates
6. Plan	Clone template or draft → search food → substitutions → constraint check → approve/publish	Inline totals; contextual search; no modal chains
7. Monitor	Attention queue: questions, missed check-ins, deviations, follow-ups	Exceptions first; bulk low-risk actions
8. Grow	Monthly theme → weekly content → approve/export → lead arrives → follow-up task	One repeatable weekly ritual

Patient journey Back to TOC

Step	Experience	Success criterion
Invite	Branded secure link, identity setup, consent, notification preference	Activation without support
Today	Current meal/goal, next action, quick question	Relevant plan visible immediately
Follow	One-tap followed/changed/skipped; optional photo/note	≤10 seconds for simple check-in
Substitute	Contextual approved alternatives with equivalent portion and explanation	No leaving meal or searching a master list
Ask	Plan-grounded answer or escalation to nutritionist	No diagnosis or silent plan change
Progress	Measurements/photos/goals and plain-language trend	Support motivation without shame or overclaim
Follow-up	Reminder and concise progress summary	Patient and professional enter return with shared context

UX principles Back to TOC

- Task-centered navigation: Consult, Plan, Monitor, Grow—not a feature warehouse.
- Autosave with visible state and version history; undo where feasible; never silently discard clinical edits.
- Patient Today view is mobile-first, thumb-reachable, WCAG-aware, low-bandwidth tolerant, and pt-BR plain language.
- Internationalization keys, locale-aware units/dates, and food-source locales from day one; no translated strings in domain logic.
- AI draft and approved content look different; uncertainty and source links are visible; rejection is one click with a reason.

18. Engineering & Testing Strategy Back to TOC

Layer	What to test	Deployment gate
Unit/domain	Nutrient calculations, portions, plan state, roles, retention, scoring rules	All changed-domain tests pass; deterministic rounding fixtures
Integration	PostgreSQL/Testcontainers, migrations, RLS/policies, outbox/jobs, signed files, auth JWT	No cross-tenant or migration regression
API/contract	OpenAPI compatibility, validation, errors, idempotency, pagination	Generated client builds; breaking change explicitly versioned
E2E	Switcher import; consult→publish; patient check-in/substitution; export; privacy request	Critical journeys pass in production-like environment
Security	SAST/SCA/secrets; IDOR/BOLA; rate limits; file upload; MFA/session; prompt injection	No unresolved critical/high; medium risk accepted in writing
Performance	p95 API; food search; plan save; bulk import; concurrent check-ins	Defined budgets met at 3× expected launch load
AI eval	Schema, groundedness, constraints, unsafe output, prompt injection, edit/acceptance	No severe safety regression; score thresholds by task
Migration/backup	Forward/backward migration rehearsal; restore DB + object references	Restore succeeds within launch RTO/RPO; checksum reconciliation
Accessibility	Keyboard, focus, screen reader, contrast, zoom, mobile touch targets	Critical flows meet WCAG 2.2 AA target

Production blockers Back to TOC

Block deployment for failed critical E2E, cross-tenant access, data loss/corruption, invalid migration, critical/high security vulnerability, unreviewed schema change, broken restore, missing rollback, severe AI safety regression, missing consent/audit on a changed regulated flow, or monitoring/alerting not receiving a test signal.

19. Product & Technical Metrics Back to TOC

North-star / metric	Definition	Initial target
Weekly successful care loops	Patient receives plan/check-in → professional sees signal → professional action when needed	Establish baseline; ≥60% active patients/week by day 90
Nutritionist activation	Imported/created patient + published plan + patient activated within 7 days	≥60% of qualified trials
Time to first patient	Signup → usable patient record	Median <30 min for supported switch import
Time to first plan	Signup → first published plan	Median <48h; in-workflow build <15 min
Patient activation	Invite → consent/account → plan viewed	≥75% within 72h
Adherence engagement	Active patients with ≥1 intended weekly check-in	≥60%, segmented by protocol
Trial conversion	Qualified trial → paid	≥20% initially; ≥35% after onboarding iteration
Nutritionist retention	Logo retention / 90-day	>85% for activated accounts
AI acceptance	Draft accepted with minor edits / reviewed outputs	Track by task; >60% only after safety threshold
AI safety	Severe constraint/clinical-policy violations	0 patient-exposed severe incidents
Availability	Successful synthetic/API checks	MVP target 99.5%; error budget reviewed
Latency/error	p95 read/write; 5xx	p95 reads <500ms, writes <800ms; 5xx <1%
Jobs	Failed/stuck job rate	<0.5%; no stuck critical jobs >15 min
AI latency/cost	p95 response; spend by task/tenant	Professional draft p95 <20s; budget hard cap
DB/backup	Connections, slow queries, restore success	No pool exhaustion; quarterly restore pass

20. Architecture Decision Records Back to TOC

ADR-001 — Frontend framework Back to TOC

Context	Multiple web clients and future mobile; solo React/TS expertise.
Options	React SPA, Next.js, Remix.
Decision	Next.js App Router + TypeScript.
Why	Strong full-stack web DX, routing, PWA path, previews; no requirement that backend live in Next.
Trade-offs	Framework/server features can increase Vercel coupling; keep API authoritative.
Revisit when	SSR/hosting economics or framework stability materially changes.

ADR-002 — Backend Back to TOC

Context	Clinical transactions, security, AI jobs, and founder Java expertise.
Options	Spring Boot, NestJS, Go.
Decision	Java 25 LTS + Spring Boot 4.1 modular monolith.
Why	Highest founder leverage; mature validation/security/transactions/observability; Spring Modulith boundaries.
Trade-offs	Two language ecosystems; more memory than Node.
Revisit when	Team composition changes or measured compute economics dominate developer productivity.

ADR-003 — Database Back to TOC

Context	Relational clinical records, versions, tenancy, audit, food data.
Options	PostgreSQL, MySQL, MongoDB.
Decision	PostgreSQL.
Why	Constraints, transactions, JSON/search extensions, portability, mature tooling.
Trade-offs	Schema discipline and migrations required.
Revisit when	A specialized workload has measured needs PostgreSQL cannot meet.

ADR-004 — Cloud/provider Back to TOC

Context	R$1,000/month ceiling, solo ops, Brazil-first latency/data location.
Options	AWS-only, GCP-only, Railway/Render, Vercel+Fly+Supabase.
Decision	Vercel Pro + Fly.io GRU + Supabase São Paulo.
Why	Best early DX/cost/region balance with managed data plane.
Trade-offs	Several vendors and public TLS DB path; no single SLA.
Revisit when	Enterprise procurement, private networking, uptime, or R$10k+ MRR justifies consolidation.

ADR-005 — Authentication Back to TOC

Context	Web/mobile identity, MFA, recovery, invitations.
Options	Build it, Keycloak, Auth0/Clerk, Cognito, Supabase Auth.
Decision	Supabase Auth; TOTP for professionals; own authorization domain.
Why	Low operational burden and regional project option; supports PKCE/MFA.
Trade-offs	Provider dependency; recovery limitations must be designed.
Revisit when	Enterprise SSO/SCIM or cost/security requirements exceed offering.

ADR-006 — API Back to TOC

Context	Multiple first-party clients and future integrations.
Options	REST, GraphQL, tRPC.
Decision	REST + OpenAPI; generated clients; webhook later.
Why	Clear contracts, caching/idempotency, mobile and third-party friendly.
Trade-offs	Potential over/under-fetching; version discipline.
Revisit when	Client composition becomes genuinely dynamic and GraphQL benefits exceed authorization cost.

ADR-007 — Multi-tenancy Back to TOC

Context	Solo practices now; clinics later; low cost.
Options	Shared schema, per-schema, per-database.
Decision	Shared DB/shared schema with organization_id + policy tests + RLS defense.
Why	Fastest/cheapest; easy migration and analytics.
Trade-offs	Highest blast radius if authorization is wrong.
Revisit when	Enterprise isolation/residency or noisy-neighbor evidence requires dedicated DB.

ADR-008 — Mobile Back to TOC

Context	Patient mobile is strategic but not in 16-week MVP.

Options	PWA, Expo/RN, Flutter, native.
Decision	Responsive PWA MVP; Expo/React Native post-MVP.
Why	Validates journeys now; later shares TS client/schemas/tokens with web.
Trade-offs	PWA push/offline/store presence limitations.
Revisit when	Patient retention proves native value and team can support releases.

ADR-009 — AI abstraction Back to TOC

Context	Models/prices/safety change; task-specific routing needed.
Options	Direct SDK, multi-provider from day 1, internal gateway.
Decision	One provider behind AiGateway + task registry/model snapshots.
Why	Avoids premature multi-provider complexity while containing coupling.
Trade-offs	Gateway and evaluation infrastructure add work.
Revisit when	Provider incident, compliance, quality, or economics requires second provider.

ADR-010 — Background processing Back to TOC

Context	Imports, exports, reminders, AI, emails must be reliable.
Options	Inline, Postgres jobs/outbox, Redis, managed queue/Kafka.
Decision	Transactional outbox + JobRunr/Postgres.
Why	Atomic with domain data and zero new infrastructure.
Trade-offs	DB polling/throughput ceiling.
Revisit when	Sustained queue load, independent scaling, or blast-radius isolation is measured.

ADR-011 — File storage Back to TOC

Context	Photos, PDFs, imports, exports; sensitive/private.
Options	DB BLOB, local disk, object storage.
Decision	Private managed object storage with signed URLs and metadata in PostgreSQL.

Why	Scalable, lifecycle-friendly, client upload; DB stays lean.
Trade-offs	Backup/restore spans systems; URL policy risk.
Revisit when	Volume/compliance needs dedicated S3 account, immutability, or regional replication.

21. Risk Register Back to TOC

Risk	Probability	Impact	Mitigation
MVP too broad for one founder	High	Critical	Freeze must-haves; weekly scope burn; growth is lite; explicit stretch labels; production at week 16 over feature count.
Fewer than five pilots produce weak evidence	High	High	Recruit 8–12 interview prospects in Sprint 0 to retain 5 active pilots; target switcher communities and churn signals.
Migration formats unavailable/inconsistent	High	High	Start with one supported CSV template + document archive; mapping preview; concierge; do not promise universal import.
Food data license/quality issue	Medium	Critical	TACO legal confirmation; visible provenance; TBCA written permission; source/version tests; custom foods.
Cross-tenant data exposure	Medium	Critical	Organization-scoped repositories, relationship policies, RLS, negative integration tests, security review, audit/alerts.
AI unsafe/hallucinated advice	High	Critical	Deterministic rules, grounding, structured output, approval gates, refusal/escalation, eval/red-team, kill switch.
Prompt injection/data exfiltration	Medium	Critical	Untrusted-content boundaries, no open tools, allowlists, output filters, minimal data, adversarial tests.
LGPD/CFN non-compliance	Medium	Critical	Counsel before beta; DPA/data map/RIPD decision; consent/retention/incident workflow; CFN marketing/telenutrition review.
Provider outage or lock-in	Medium	High	Postgres/OpenAPI/OIDC/storage adapters; backups/export; containers; provider status alerts; recovery playbook.
Cost exceeds R$1,000	Medium	High	Hard AI/provider budgets, alerts at 50/75/90%, quotas, no PITR initially, free telemetry tiers, monthly unit economics.
Incumbents copy AI/growth features	High	Medium	Compete on integrated workflow, migration, UX reliability, data feedback, governance, and customer intimacy.

Risk	Probability	Impact	Mitigation
Patient engagement remains low	Medium	High	Today-first design; configurable low-friction check-ins; onboarding tests; notification consent/cadence; measure activation.
Nutritionists reject AI as commoditizing profession	Medium	High	Professional-control positioning, labels, audit, edit/approval, explain value as time reclaimed and better continuity.
Social content violates ethics	Medium	High	Ruleset/checklist, approval, evidence, forbidden-pattern tests, no auto-post; professional remains responsible.
Solo-founder support overload	High	High	Narrow supported migration, in-product diagnostics, admin tools, runbooks, office hours, capped founding cohort.
Retention/deletion conflict	Medium	High	Purpose-based schedule, legal holds, deactivation vs deletion, export, counsel-validated workflow, audit.

22. Sprint-by-Sprint Implementation Plan Back to TOC

Critical path

Pilot recruitment → migration contract/sample exports → tenancy/security foundation → food provenance and plan builder → patient publication/check-in → real pilot feedback → LGPD/CFN/legal validation → restore/security/AI gates → paid production.

Sprint 0 (weeks 1–2) — Discovery & Technical Foundation Back to TOC

Objective	Discovery & Technical Foundation
Features	Interview at least 5 switchers; recruit 8–12 prospects; map current exports and plan creation; clickable prototypes for import, plan builder, Today, and Growth; define beta agreement.
Technical work	Monorepo/repositories; Java/Spring and Next skeletons; CI; environments; ADR baseline; data map/threat model; OpenAPI; telemetry; Supabase/Fly/Vercel spikes; TACO/TBCA legal inquiry.
Deliverables	Validated problem brief; ranked JTBD; prototype findings; supported-import contract; architecture runway; measurement plan; pilot roster.
Acceptance criteria	≥5 completed interviews including ≥3 active incumbent users; ≥3 agree to pilot; median prototype task completion; no unresolved architecture blocker; scope signed off.
Dependencies	Access to nutritionists; sample exports; legal/privacy consultation scheduling.
Complexity	XL

Sprint 1 (weeks 3–4) — Identity, Tenancy & Switcher Onboarding Back to TOC

Objective	Identity, Tenancy & Switcher Onboarding
Features	Professional signup/profile; organization; patient invitation shell; CSV import mapping/preview/errors; document archive upload; export skeleton.
Technical work	Supabase Auth/PKCE/TOTP; membership/RBAC/relationship policies; tenant-scoped repositories/RLS; Flyway; signed file uploads; audit; import job/outbox; fake-data admin.
Deliverables	Deployed staging; import supported template; tenant-security test suite; onboarding analytics.
Acceptance criteria	Two organizations cannot access each other by API/object URL; import is idempotent/reversible pre-commit; error report identifies every skipped row; MFA works for professional.
Dependencies	Sprint 0 import contract; identity/provider account; DPA draft.
Complexity	XL

Sprint 2 (weeks 5–6) — Patient Record, Intake & Consultation Back to TOC

Objective	Patient Record, Intake & Consultation
Features	Patient profile/care relationship; consent; intake/anamnesis; core clinical history; consultation; typed notes; goals; core measurements/photos.
Technical work	Versioned clinical notes; private media metadata; validation; autosave; audit; privacy-request/export data model; accessibility baseline.
Deliverables	End-to-end create/import patient → intake → consultation flow; tested prototypes with pilots.
Acceptance criteria	Autosave survives refresh; final note immutable with amendment; consent evidence versioned; imported and newly created patients behave consistently; zero critical pilot usability defects.
Dependencies	Tenancy/auth; consent/legal wording; pilot availability.
Complexity	XL

Sprint 3 (weeks 7–8) — Food Data & Fast Plan Builder Back to TOC

Objective	Food Data & Fast Plan Builder
Features	TACO food search/provenance; custom foods; meals/items/portions; nutrient totals; template clone; substitutions; plan draft/review/publish/supersede; PDF.

Technical work	Food ingestion/normalization tests; deterministic calculators/rounding; plan aggregate/versioning; optimistic locking/autosave; search indexes; constraint engine v1.
Deliverables	Professional can build and publish a standard plan; patient-ready API; calculation reference fixtures.
Acceptance criteria	Reference totals match fixtures; provenance visible; concurrent edit conflict is explicit; patient never sees draft; trained median plan build ≤15 min in target scenario.
Dependencies	Food legal decision; consultation/patient model; UX tests.
Complexity	XL — highest delivery risk

Sprint 4 (weeks 9–10) — Patient PWA & Adherence Loop Back to TOC

Objective	Patient PWA & Adherence Loop
Features	Invite/onboarding; Today; plan/recipe-context placeholder; approved substitutions; one-tap check-in; measurements/photos; question inbox; email reminders; progress graphs.
Technical work	Mobile-responsive PWA; notification preference; engagement events; attention rules; signed media; rate limits; synthetic patient journey; sanitized analytics.
Deliverables	Real pilot patient can receive plan, follow/check in, substitute, ask, and see progress; nutritionist sees attention queue.
Acceptance criteria	Patient simple check-in ≤10 sec; no cross-patient access; reminders honor preference; attention flag contains explainable evidence; WCAG critical-flow review passes.
Dependencies	Published plan API; email domain; pilot patients/consent.
Complexity	XL

Sprint 5 (weeks 11–12) — AI Copilot & Private Beta Back to TOC

Objective	AI Copilot & Private Beta
Features	Pre-consult summary; typed notes → structured draft; professional plan draft; adherence digest; plan-grounded patient Q&A behind flag; approval/rejection; usage display.
Technical work	AiGateway/task registry; store:false; structured outputs; prompt/policy/model versions; PHI minimization; cost/latency telemetry; eval harness; injection defenses; kill switches.
Deliverables	Private beta release; AI evaluation report; review/audit UI; per-tenant budget controls.
Acceptance criteria	Schema-valid ≥99% on eval; zero patient-exposed severe policy violations in golden/red-team set; every output traceable; budget cap works; beta legal gates signed.

Dependencies	Core data loop; AI DPA/transfer review; nutritionist evaluators.
Complexity	XL

Sprint 6 (weeks 13–14) — Practice Growth Lite & Commercial Onboarding Back to TOC

Objective	Practice Growth Lite & Commercial Onboarding
Features	Positioning brief; monthly/weekly content plan; caption/video outline drafts; evidence/ethics checklist; approval/export; public professional profile + lead form; follow-up task; founding plan administration.
Technical work	Separate growth/lead domain; anti-spam/rate limit; content prompt/eval; prohibited-pattern checks; conversion events; secure lead-to-patient conversion; hosted payment link/manual entitlement.
Deliverables	One complete growth ritual and lead pipeline; pricing/trial screen; switcher onboarding guide.
Acceptance criteria	No draft auto-publishes; forbidden fee/promotion/before-after scenarios flagged; lead data separated from clinical record; ≥3 pilots complete weekly planning; commercial entitlement auditable.
Dependencies	CFN marketing review; AI foundation; landing/profile copy.
Complexity	L

Sprint 7 (weeks 15–16) — Hardening, Migration Pilot & Production Launch Back to TOC

Objective	Hardening, Migration Pilot & Production Launch
Features	Pilot fixes; migration concierge; privacy center; admin/support tools; status/help; launch onboarding; feedback and cancellation/export.
Technical work	Load/security/accessibility tests; backup + object restore drill; incident tabletop; SLO/alerts; log scrub; dependency scan; rollback; production checklist; runbooks; cost alarms.
Deliverables	Production release; first paid switchers; launch report; prioritized post-MVP backlog.
Acceptance criteria	All production blockers clear; RPO/RTO drill passes; no unresolved critical/high security findings; ≥3 real practices migrated; ≥2 paid/committed switchers; costs project ≤R$1,000; rollback tested.
Dependencies	All prior sprints; counsel sign-off; pilot availability; provider production accounts.
Complexity	XL

Scope contingency Back to TOC

If the plan builder or patient loop slips, cut in this order: public profile/lead form → content video variants → patient AI Q&A → progress photos → document batch convenience. Do not cut tenant isolation, exports, consent/audit, plan versioning, deterministic calculations, patient Today/check-in, backups, or release gates.

23. MVP Launch Checklist Back to TOC

Product and pilots Back to TOC

- ☐ At least three switcher practices migrated and using the full loop with real patients.
- ☐ Activation, time-to-plan, patient activation, adherence, and cancellation/export events visible.
- ☐ MVP scope/help/pricing/trial/support boundaries published in pt-BR.
- ☐ No critical pilot issue without owner, workaround, and deadline.

Clinical and AI safety Back to TOC

- ☐ Nutritionist approval gates verified; draft/approved labels visible; no autonomous plan modification.
- ☐ Constraint/calculation fixtures and AI evaluation thresholds pass; kill switches tested.
- ☐ Patient escalation and emergency disclaimer reviewed; no open-web clinical answering.

Privacy, legal, and professional rules Back to TOC

- ☐ Terms, privacy notice, DPA, subprocessors, data-flow map, lawful-basis/retention schedule reviewed by counsel.
- ☐ Consent, guardian path decision, privacy requests, export, withdrawal, deletion/legal-hold flows tested.
- ☐ CFN/CRN review of telenutrition boundary, professional documents, marketing assistant, and record retention complete.
- ☐ International-transfer mechanism and AI/provider configurations documented.

Security and reliability Back to TOC

- ☐ Threat model closed; tenant/IDOR/object tests pass; MFA and recovery verified.
- ☐ Private files, signed URLs, upload validation/scanning, secrets, dependency/container scans pass.
- ☐ Daily backups, object inventory, restore, rollback, incident tabletop, alerts, and status communication tested.
- ☐ Logs/analytics/AI telemetry sampled and confirmed free of clinical payloads/secrets.

Operations and economics Back to TOC

- ☐ Provider and AI hard budgets/alerts configured; forecast ≤R$1,000 for first six months.
- ☐ Support/admin/runbooks, incident contacts, migration playbook, cancellation/export procedure ready.
- ☐ Hosted billing/manual entitlements, invoice/refund/tax/accounting process ready.
- ☐ Founder on-call capacity and founding-cohort cap explicitly set.

24. Open Questions / Decisions Required Back to TOC

Decision	Recommended default	Deadline / consequence
Exact switcher segment	Independent clinical nutritionists with 20–150 active patients	Sprint 0; changes protocols, acquisition, and pricing
Pilot recruitment	Recruit 8–12 prospects to retain ≥5 active; at least 3 current incumbent users	Week 1; fewer than 3 active pilots makes week-16 paid launch high-risk
Supported migration source	Start with a canonical CSV + document archive; add one competitor export only after samples	Week 2; defines onboarding promise
Food data permission	Use TACO only after confirmation; request written TBCA commercial/bulk permission	Before Sprint 3; can block plan builder
Telenutrition scope	Do not include video; record remote modality/consent if appointments represent telenutrition	Before beta; legal/CFN workflow impact
Children/adolescents	Exclude under-18s from beta unless guardian/consent/clinical workflow is designed and reviewed	Before beta; material LGPD/clinical risk
AI patient assistant	Feature-flag controlled beta; approved plan only; no autonomous modifications	Sprint 5 gate; can be cut without breaking core MVP
Social-growth claims	CFN-reviewed rule pack; no auto-post, fees/promotions, before-after, guarantees, or brand endorsements	Before Sprint 6 pilot
Data roles and retention	Nutritionist controller / SaaS operator for care data; purpose-based separate controller roles	Counsel before real data
Brand/name/domain	Decide after 5–8 switcher interviews and positioning tests	By week 4; avoid naming driving scope

Immediate next action

Do not start the production plan builder first. In week 1, obtain real export samples from at least two switchers, observe one complete consultation-to-plan workflow, test the migration/Today prototypes, and book Brazilian privacy + CFN-specialist review. Those inputs determine whether the 16-week plan is credible.

Appendix A — Principal Sources and Research Notes Back to TOC

Research performed 18 August 2026. Official sources are preferred. App-store/community evidence is observational. Pricing may change and should be re-checked before launch.

Source: WebDiet — product

Source: WebDiet — pricing

Source: WebDiet — privacy

Source: WebDiet patient app

Source: Dietbox — product/pricing

Source: Dietbox patient app

Source: Dietbox professional App Store reviews

Source: Nutrium

Source: Nutrium client export
Source: NutriAssist
Source: Salutes
Source: Healthie pricing
Source: Practice Better pricing
Source: That Clean Life pricing
Source: NutriAdmin pricing
Source: TACO
Source: TBCA
Source: LGPD
Source: ANPD incidents
Source: ANPD international transfer
Source: CFN ethics
Source: CFN telenutrition
Source: CFN records
Source: OpenAI models
Source: OpenAI data controls
Source: Supabase pricing
Source: Supabase regions
Source: Fly.io regions
Source: Fly.io pricing
Source: Vercel pricing
Source: PostHog pricing